Drift-Triggered Retraining with an Orchestrated Serverless Endpoint

Short project report | simulated churn drift across eight weeks

1. Objective and method

This project trains a reproducible Random Forest churn model, registers versions with validation metrics and SHA-256 checksums, and simulates eight sequential weeks of data. Population Stability Index (PSI) measures feature drift: below 0.10 is low, 0.10–0.25 moderate, and ≥0.25 high. Weekly performance is measured separately on labelled data for retrospective analysis. Promotion decisions use current-week features plus validation on reference/prior labelled data only; current/future week labels are not used to decide promotion for that week.

2. Findings

Baseline model weekly retrospective F1 ranges from 0.271 to 0.504. Mean feature PSI rises later in the simulation: week 6 0.310, week 7 0.501, week 8 0.341. Drift and performance loss are therefore assessed separately; drift alone is not treated as proof of poor predictions.

Policy comparison

Policy	Attempts	Promotions	Mean weekly F1	Cost units
drift_high	4	4	0.401	48
drift_low	5	5	0.402	60
never	0	0	0.383	0
weekly	8	5	0.401	90

Assumptions: 10 abstract cost units per retraining attempt and 2 per successful promotion (not cloud billing). The high-threshold policy had mean F1 0.401 at cost 48; low-threshold had 0.402 at cost 60. The high threshold costs 20% less for a very small mean-F1 difference, so it is recommended as the economical balance on this dataset.

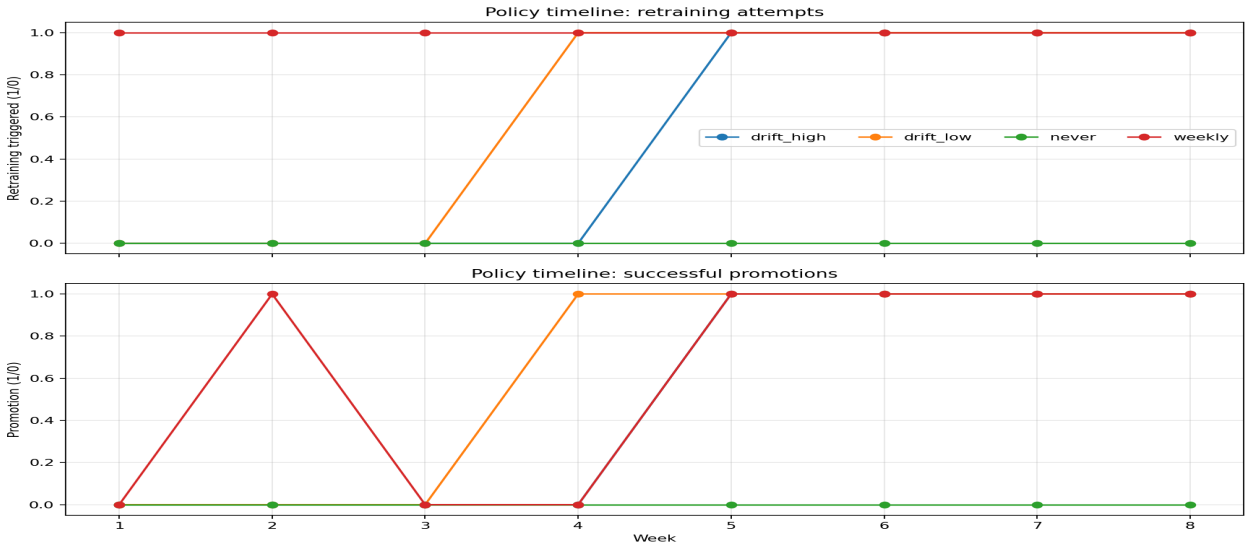


Figure 1. Retraining attempts and successful promotions by week and policy.

3. Safety, retry and escalation

Fault evidence covers invalid schema (run fails closed), forced candidate rejection (no promotion), and a transient ingestion timeout (two retries, then success). The invalid-data and rejected-candidate cases leave the registry production version and endpoint-resolved version unchanged. Retries are bounded to two with exponential backoff and apply only to transient timeout/connection failures; schema faults are not retried.

Escalation uses current-week feature data only: trigger if at least two features have PSI ≥0.25, or mean PSI ≥0.25. It fired in weeks 6, 7, 8 in the final run. The run history stores PSI values and the reason.

4. Endpoint, reproducibility and limitations

The notebook measured 30 local cold model-load calls (mean 34.3 ms) and 30 warm cached calls (mean 10.7 ms) in the preparation environment. These are local timings, not hosted cloud cold starts. The project includes a PC-runnable FastAPI endpoint and an AWS Lambda-compatible handler/template. Actual AWS deployment requires an AWS account, permissions, packaging, and separate provider-side cold/warm measurements; it is not claimed as completed here. Run the notebook top-to-bottom to regenerate registry evidence, the eight-week history, drift charts, workflow graph, policy table/timeline, fault logs, retry evidence and endpoint-switch proof. AI assistance was used for planning, scaffolding and documentation; the team must review and verify the work.